Classical theories of information science

率失真理论+AI

涂仕奎
上海交通大学
tushikui@sjtu.edu.cn
https://www.cs.sjtu.edu.cn/~tushikui/

Outline

1. The Rate Distortion Theory
2. The Information Bottleneck Theory
3. Why Language Models Hallucinate?

1. The Rate Distortion Theory

Information Transmission with Distortion

- Consider compressing an information source with entropy rate H at rate $R < H$.
- By the source coding theorem, $P_e \to 1$ as $n \to \infty$.
- Under such a situation, information must be transmitted with "distortion".
- What is the best possible tradeoff?

Single-Letter Distortion Measure

- Let $\{X_k, k \geq 1\}$ be an i.i.d. information source with generic random variable $X \sim p(x)$, where $|\mathcal{X}| < \infty$.
- Consider a source sequence $\mathbf{x} = (x_1, x_2, \cdots, x_n)$ and a reproduction sequence $\hat{\mathbf{x}}$

Definition 8.1 A single-letter distortion measure is a mapping

$$d : \mathcal{X} \times \hat{\mathcal{X}} \to \Re^+.$$

where

- The con
- $|\hat{\mathcal{X}}| < \infty$

The value $d(x, \hat{x})$ denotes the distortion incurred when a source symbol x is reproduced as $\hat{x}$.

- In general, $\hat{\mathcal{X}}$ may be different from $\mathcal{X}$.
- For example, $\hat{\mathbf{x}}$ can be a quantized version of $\mathbf{x}$.

Single-Letter Distortion Measure

Definition 8.1 A single-letter distortion measure is a mapping

$$d : \mathcal{X} \times \hat{\mathcal{X}} \to \Re^+.$$

The value $d(x, \hat{x})$ denotes the distortion incurred when a source symbol x is reproduced as $\hat{x}$.

Definition 8.2 The average distortion between a source sequence $\mathbf{x} \in \mathcal{X}^n$ and a reproduction sequence $\hat{\mathbf{x}} \in \hat{\mathcal{X}}^n$ induced by a single-letter distortion measure d is defined by

$$d(\mathbf{x}, \hat{\mathbf{x}}) = \frac{1}{n} \sum_{k=1}^{n} d(x_k, \hat{x}_k).$$

Examples of a Distortion Measure

- Let $\hat{\mathcal{X}} = \mathcal{X}$.

  1. Square-error: $d(x, \hat{x}) = (x - \hat{x})^2$, where $\mathcal{X}$ and $\hat{\mathcal{X}}$ are real.
  2. Hamming distortion:

$$d(x, \hat{x}) = \begin{cases} 0 & \text{if } x = \hat{x} \\ 1 & \text{if } x \neq \hat{x} \end{cases}$$

  where the symbols in $\mathcal{X}$ do not carry any particular meaning.

Examples of a Distortion Measure

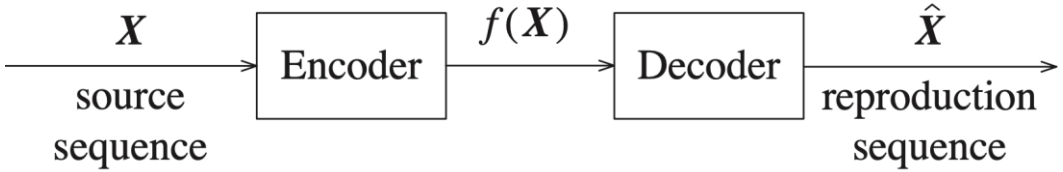
- Let $\hat{X}$ be an estimate of X.

  1. If d is the square-error distortion measure, $Ed(X, \hat{X})$ is called the mean square error.
  2. If d is the Hamming distortion measure,

$$Ed(X, \hat{X}) = \Pr\{X = \hat{X}\} \cdot 0 + \Pr\{X \neq \hat{X}\} \cdot 1 = \Pr\{X \neq \hat{X}\}$$

  is the probability of error. For a source sequence $\mathbf{x}$ and a reproduction sequence $\hat{\mathbf{x}}$, the average distortion $d(\mathbf{x}, \hat{\mathbf{x}})$ gives the frequency of error in $\hat{\mathbf{x}}$.

The Rate-Distortion Code



All the discussions are with respect to an i.i.d. information source $\{X_k, k \geq 1\}$ with generic random variable X and a distortion measure d.

Definition 8.8 An (n, M) rate-distortion code is defined by an encoding function

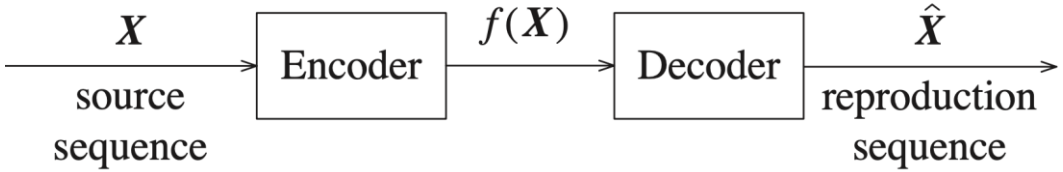
$$f : \mathcal{X}^n \to \{1, 2, \cdots, M\}$$

and a decoding function

$$g : \{1, 2, \cdots, M\} \to \hat{\mathcal{X}}^n.$$

The set $\{1, 2, \cdots, M\}$, denoted by $\mathcal{I}$, is called the index set. The reproduction sequences $g(1), g(2), \cdots, g(M)$ in $\hat{X}^n$ are called codewords, and the set of codewords is called the codebook.

The Rate-Distortion Code



Definition 8.9 The rate of an (n, M) rate-distortion code is $n^{-1} \log M$ in bits per symbol.

Definition 8.10 A rate-distortion pair (R, D) is (asymptotically) achievable if for any $\epsilon > 0$, there exists for sufficiently large n an (n, M) rate-distortion code such that

$$\frac{1}{n} \log M \leq R + \epsilon$$

and

$$\Pr\{d(\mathbf{X}, \hat{\mathbf{X}}) > D + \epsilon\} \leq \epsilon,$$

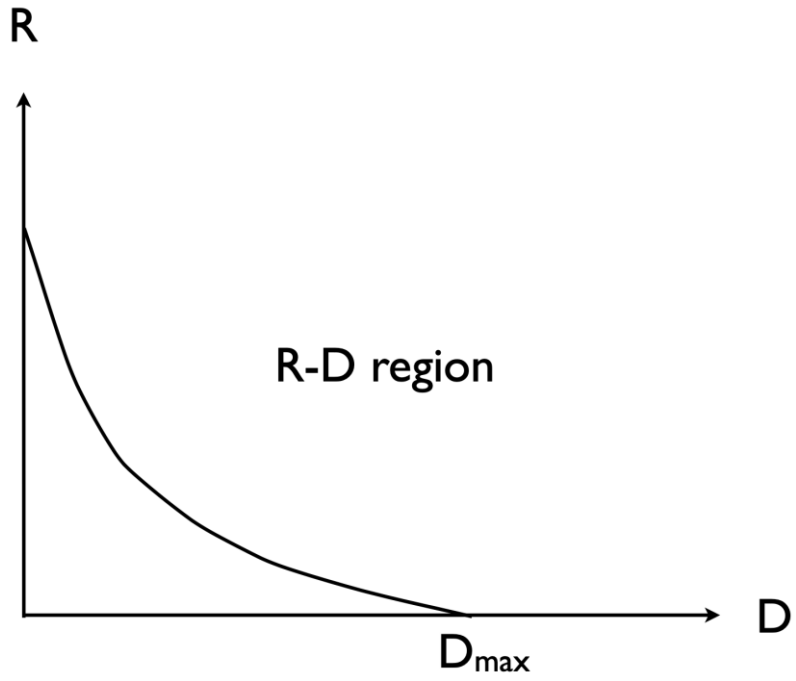
where $\hat{\mathbf{X}} = g(f(\mathbf{X}))$.

Remark If (R, D) is achievable, then (R', D) and (R, D') are achievable for all $R' \geq R$ and $D' \geq D$. This in turn implies that (R', D') are achievable for all $R' \geq R$ and $D' \geq D$.

The Rate Distortion Function

Definition 8.13 The rate-distortion function $R(D)$ is the minimum of all rates R for a given distortion D such that (R, D) is achievable.

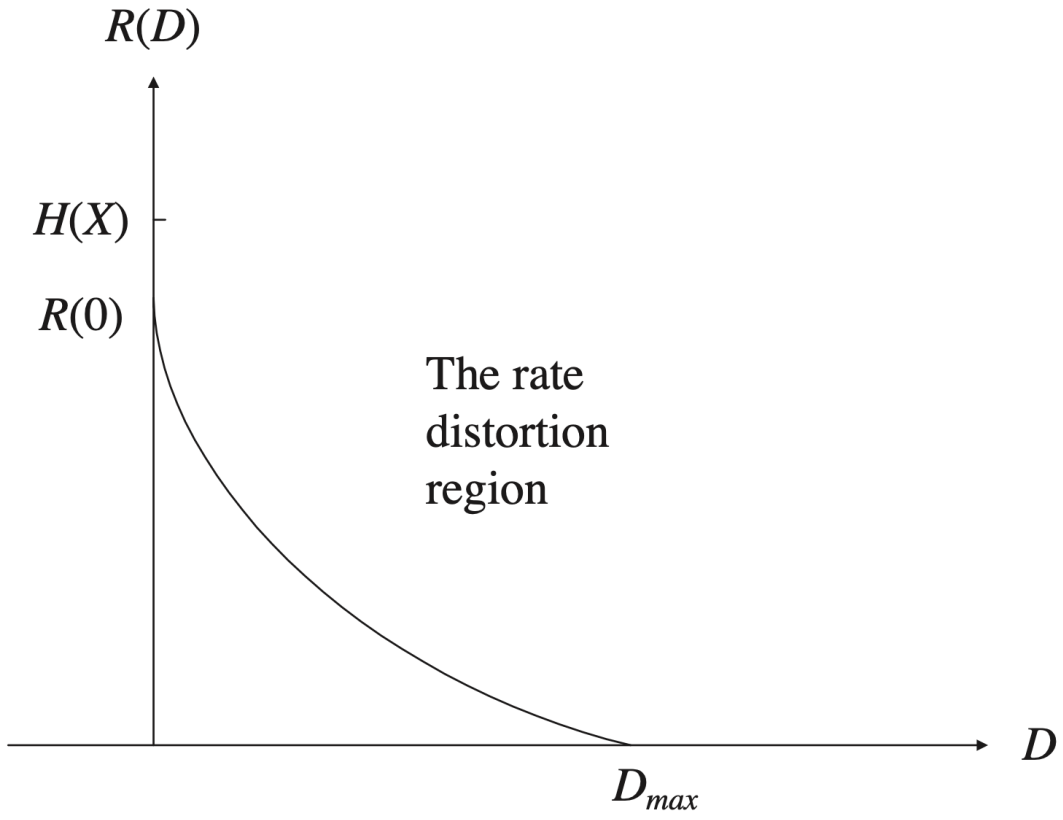
Definition 8.14 The distortion-rate function $D(R)$ is the minimum of all distortions D for a given rate R such that (R, D) is achievable.



The Rate Distortion Properties

Theorem 8.15 The following properties hold for the rate-distortion function $R(D)$:

1. $R(D)$ is non-increasing in D.
2. $R(D)$ is convex.
3. $R(D) = 0$ for $D \geq D_{max}$.
4. $R(0) \leq H(X)$.



Proof

1. Let $D' \geq D$. $(R(D), D)$ achievable $\Rightarrow$ $(R(D), D')$ achievable. Then $R(D) \geq R(D')$ by definition of $R(\cdot)$.
2. Follows from the convexity of the rate-distortion region.
3. $(0, D_{max})$ is achievable $\Rightarrow$ $R(D) = 0$ for $D \geq D_{max}$.
4. Since d is assumed to be normal, $(H(X), 0)$ is achievable, and hence $R(0) \leq H(X)$.

The Rate Distortion Theorem

Definition 8.16 For $D \geq 0$, the information rate-distortion function is defined by

$$R_I(D) = \min_{\hat{X}: Ed(X,\hat{X}) \leq D} I(X;\hat{X}).$$

- The minimization is taken over the set of all $p(\hat{x}|x)$ such that $Ed(X,\hat{X}) \leq D$ is satisfied, namely the set

$$\left\{ p(\hat{x}|x) : \sum_{x,\hat{x}} p(x)p(\hat{x}|x)d(x,\hat{x}) \leq D \right\}.$$

- Since this set is compact in $\Re^{|\mathcal{X}||\hat{\mathcal{X}}|}$ and $I(X;\hat{X})$ is a continuous functional of $p(\hat{x}|x)$, the minimum value of $I(X;\hat{X})$ can be attained.

The Rate Distortion Theorem

Definition 8.16 For $D \geq 0$, the information rate-distortion function is defined by

$$R_I(D) = \min_{\hat{X}: Ed(X,\hat{X}) \leq D} I(X;\hat{X}).$$

Theorem 8.17 (The Rate-Distortion Theorem) $R(D) = R_I(D)$.

Theorem 8.18 The following properties hold for the information rate-distortion function $R_I(D)$:

1. $R_I(D)$ is non-increasing in D.
2. $R_I(D)$ is convex.
3. $R_I(D) = 0$ for $D \geq D_{max}$.
4. $R_I(0) \leq H(X)$.

两个经典例子

Example 1：Binary i.i.d. source, Hamming distortion

$X \sim \text{Bernoulli}(p)$，$p \le 0.5$，汉明失真 $d(x,\hat{x}) = \mathbb{I}(x \neq \hat{x})$

$$R(D) = \begin{cases} H(p) - H(D), & 0 \le D \le p \\ 0, & D > p \end{cases}$$

$H(\cdot)$：二元熵函数。

- $D = 0$：$R(0) = H(p)$，回到无损信源编码定理；
- $D = p$：$R(p) = 0$，直接固定输出 0，不需要传比特。

Example 2：Zero-mean Gaussian source, squared-error distortion

$X \sim \mathcal{N}(0, \sigma^2)$，$d(x,\hat{x}) = (x - \hat{x})^2$

$$R(D) = \begin{cases} \frac{1}{2}\log_2 \frac{\sigma^2}{D}, & 0 < D \le \sigma^2 \\ 0, & D > \sigma^2 \end{cases}$$

> 这就是著名的 **Gaussian rate-distortion law**，多媒体有损压缩的理论基准。

Rate Distortion Theorem vs Shannon’s other theorems

Theorem	Optimization problem	Goal
Noiseless Source Coding	$\min R,\ \text{error} \to 0$	无损压缩下界 $H(X)$
Channel Coding	$\max C,\ \text{error} \to 0$	信道可靠传输最大码率
Rate Distortion	$\min R,\ \mathbb{E}[d] \leq D$	有损压缩下界 $R(D)$

> 对偶关系：
>
> Channel capacity: $C = \max_{p(x)} I(X;Y)$
>
> Rate distortion: $R(D) = \min_{p(\hat{x}|x)} I(X;\hat{X})$
>
> 二者都可以用 Blahut-Arimoto 迭代数值求解。

2. The Information Bottleneck Theory

The paper

- 信息瓶颈（IB）方法由 Tishby、Pereira 和 Bialek 于 2000 年提出，解决了信息论和机器学习中的一个基本问题：如何仅从数据中提取相关信息。
- 虽然香农的经典信息论提供了量化和传输信息的工具，但它本身并没有解决相关性的概念。
- IB 方法通过将有意义信息的提取形式化为信息论优化问题来填补这一空白。
- 信息瓶颈方法的核心是，相关信息可以通过原始信号和单独的“相关性变量”之间的关系来定义。 这种方法不需手动指定失真函数，在实际场景中通常难以适当地定义失真函数。

The information bottleneck method

Naftali Tishby,[1,2] Fernando C. Pereira,[3] and William Bialek[1]

[1]NEC Research Institute, 4 Independence Way
Princeton, New Jersey 08540
[2]Institute for Computer Science, and
Center for Neural Computation
Hebrew University
Jerusalem 91904, Israel
[3]AT&T Shannon Laboratory
180 Park Avenue
Florham Park, New Jersey 07932

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The main idea

- 信息瓶颈方法扩展了率失真理论，该理论传统上解决有损压缩问题。 在经典的率失真理论中，预定义的失真函数衡量压缩质量。 然而，为复杂数据定义合适的失真函数仍然具有挑战性。
- IB 方法的关键见解是用从数据和相关性变量之间的统计关系导出的隐式度量来代替显式失真函数。 这种方法将重点从保留任意信号属性转移到专门保留与目标变量相关的信息。

- 形式上，给定两个随机变量 X 和 Y 的联合分布 $p(x,y)$，IB 方法旨在找到 X 的压缩表示 T，该表示：
  - **最小化 T 中包含的关于 X 的信息（压缩）**
  - **最大化 T 中包含的关于 Y 的信息（相关性保留）**

- 这种双重目标创建了一个自然的“瓶颈”，迫使表示仅关注 X 中关于 Y 的信息方面。

https://www.alphaxiv.org/zh/abs/physics/0004057

Relevant quantization

- Let X denote the signal (message) space with a fixed probability measure $p(x)$, and let $\tilde{X}$ denote its quantized codebook or compressed representation.
- The mapping $p(\tilde{x}|x)$ induces a soft partitioning of X in which each block is associated with one of the codebook elements $\tilde{x} \in \tilde{X}$, with probability given by

$$p(\tilde{x}) = \sum_x p(x)p(\tilde{x}|x)$$

- The average volume of the elements of X that are mapped to the same codeword is $2^{H(X|\tilde{X})}$, where $$H(X|\tilde{X}) = -\sum_{x\in X} p(x) \sum_{\tilde{x}\in\tilde{X}} p(\tilde{x}|x) \log p(\tilde{x}|x)$$
- What determines the quality of a quantization? The first factor is of course the rate, or the average number of bits per message needed to specify an element in the codebook without confusion.

Relevant quantization

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- What determines the quality of a quantization? The first factor is of course **the rate**, or **the average number of bits per message** needed to specify an element in the codebook without confusion.
- This number per element of X is bounded from below by the mutual information: $$I(X;\tilde{X}) = \sum_{x\in X}\sum_{\tilde{x}\in\tilde{X}} p(x,\tilde{x}) \log\left[\frac{p(\tilde{x}|x)}{p(\tilde{x})}\right]$$
- since the average cardinality of the partitioning of X is given by the ratio of the volume of X to that of the mean partition $2^{H(X)}/2^{H(X|\tilde{X})} = 2^{I(X;\tilde{X})}$, via the standard asymptotic arguments.

Relevance through distortion: Rate distortion theory

- However, information rate alone is not enough to characterize good quantization since the rate can always be reduced by throwing away details of the original signal x. We need therefore some additional constraints.
- In **rate distortion theory** such a constraint is provided through a distortion function, $d: X \times \tilde{X} \rightarrow \mathcal{R}^{+}$, which is presumed to be small for good representations.
- The partitioning of X induced by the mapping $p(\tilde{x}|x)$ has an **expected distortion**:

$$\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})} = \sum_{x\in X}\sum_{\tilde{x}\in\tilde{X}} p(x,\tilde{x})d(x,\tilde{x})$$

- Tradeoff between the rate of the quantization and the expected distortion: **the larger the rate, the smaller is the achievable distortion.**

Rate distortion theory

$$\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})} = \sum_{x\in X}\sum_{\tilde{x}\in\tilde{X}} p(x,\tilde{x})d(x,\tilde{x})$$

- The celebrated rate distortion theorem characterizes this tradeoff through the **rate distortion function**, $R(D)$, defined as the minimal achievable rate under a given constraint on the expected distortion:

$$R(D) \equiv \min_{\{p(\tilde{x}|x):\langle d(x,\tilde{x})\rangle \leq D\}} I(X;\tilde{X})$$

- Finding the rate distortion function is a variational problem that can be solved by introducing a Lagrange multiplier, β, for the constrained expected distortion. One then needs to minimize the functional:

$$\mathcal{F}[p(\tilde{x}|x)] = I(X;\tilde{X}) + \beta\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})}$$

Solution

minimize $\mathcal{F}[p(\tilde{x}|x)] = I(X;\tilde{X}) + \beta\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})}$

Theorem 1 *The solution of the variational problem,*

$$\frac{\delta \mathcal{F}}{\delta p(\tilde{x}|x)} = 0, \tag{7}$$

for normalized distributions $p(\tilde{x}|x)$, is given by the exponential form

$$p(\tilde{x}|x) = \frac{p(\tilde{x})}{Z(x,\beta)} \exp\left[-\beta d(x,\tilde{x})\right] \ , \tag{8}$$

where $Z(x,\beta)$ is a normalization (partition) function. Moreover, the Lagrange multiplier β, determined by the value of the expected distortion, D, is positive and satisfies

$$\frac{\delta R}{\delta D} = -\beta \ . \tag{9}$$

Solution

minimize $\mathcal{F}[p(\tilde{x}|x)] = I(X;\tilde{X}) + \beta\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})}$

$$p(\tilde{x}|x) = \frac{p(\tilde{x})}{Z(x,\beta)} \exp\left[-\beta d(x,\tilde{x})\right] \ , \qquad (8)$$

Proof. Taking the derivative w.r.t. $p(\tilde{x}|x)$, for given x and $\tilde{x}$, one obtains

$$\frac{\delta\mathcal{F}}{\delta p(\tilde{x}|x)} = p(x)\left[\log\frac{p(\tilde{x}|x)}{p(\tilde{x})} + 1 - \frac{1}{p(\tilde{x})}\sum_{x'} p(x')p(\tilde{x}|x') + \beta d(x,\tilde{x}) + \frac{\lambda(x)}{p(x)}\right], \qquad (10)$$

since the marginal distribution satisfies $p(\tilde{x}) = \sum_{x'} p(x')p(\tilde{x}|x')$. $\lambda(x)$ are the normalization Lagrange multipliers for each x. Setting the derivatives to zero and writing $\log Z(x,\beta) = \lambda(x)/p(x)$, we obtain Eq. (8). When varying the normalized $p(\tilde{x}|x)$, the variations $\delta I(X;\tilde{X})$ and $\delta\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})}$ are linked through

$$\delta\mathcal{F} = \delta I(X;\tilde{X}) + \beta\delta\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})} = 0, \qquad (11)$$

from which Eq. (9) follows. The positivity of β is then a consequence of the concavity of the rate distortion function (see, for example, Chapter 13 of Ref. [2]). □

A Lemma

Lemma 2 *Let $p(x,y) = p(x)p(y|x)$ be a given joint distribution. Then the distribution $q(y)$ that minimizes the relative entropy or Kullback–Leibler divergence, $D_{KL}[p(x,y)|p(x)q(y)]$, is the marginal distribution*

$$p(y) = \sum_x p(x)p(y|x).$$

Namely,

$$I(X;Y) = D_{KL}[p(x,y)|p(x)p(y)] = \min_{q(y)} D_{KL}[p(x,y)|p(x)q(y)] \ .$$

Equivalently, the distribution $q(y)$ which minimizes the expected relative entropy,

$$\sum_x p(x) D_{KL}[p(y|x)|q(y)],$$

is also the marginal distribution $p(y) = \sum_x p(x)p(y|x)$.

Remarks on the Lemma

- $D_{KL}[p(x,y)\|p(x)q(y)]$ measures how far the joint distribution $p(x,y)$ is from the product distribution $p(x)q(y)$ (the distribution if X, Y were independent).
- To make $p(x)q(y)$ as close as possible to true $p(x,y)$, we must set $q(y)$ equal to the true marginal $p(y)$.
- Mutual information $I(X;Y)$ is exactly the minimal KL distance between the true joint and any product distribution $p(x)q(y)$.

The Blahut–Arimoto algorithm

$$p(\tilde{x}) = \sum_x p(x)p(\tilde{x}|x) \tag{1}$$

$$p(\tilde{x}|x) = \frac{p(\tilde{x})}{Z(x,\beta)} \exp\left[-\beta d(x,\tilde{x})\right] \ , \tag{8}$$

Theorem 3 *Equations (1) and (8) are satisfied simultaneously at the minimum of the functional,*

$$\mathcal{F} = -\langle \log Z(x,\beta) \rangle_{p(x)} = I(X;\tilde{X}) + \beta \langle d(x,\tilde{x}) \rangle_{p(x,\tilde{x})} \ , \tag{12}$$

where the minimization is done independently *over the convex sets of the normalized distributions,* $\{p(\tilde{x})\}$ *and* $\{p(\tilde{x}|x)\}$,

$$\min_{p(\tilde{x})} \min_{p(\tilde{x}|x)} \mathcal{F}\left[p(\tilde{x}); p(\tilde{x}|x)\right] \ .$$

These independent conditions correspond precisely to alternating iterations of Eq. (1) and Eq. (8). Denoting by t *the iteration step,*

$$\begin{cases} p_{t+1}(\tilde{x}) = \sum_x p(x) p_t(\tilde{x}|x) \\ p_t(\tilde{x}|x) = \frac{p_t(\tilde{x})}{Z_t(x,\beta)} \exp(-\beta d(x,\tilde{x})) \end{cases} \tag{13}$$

where the normalization function $Z_t(x,\beta)$ *is evaluated for every* t *in Eq. (13). Furthermore, these iterations converge to a unique minimum of* $\mathcal{F}$ *in the convex sets of the two distributions.*

Relevance through another variable: The Information Bottleneck

- Since the "right" distortion measure is rarely available, the problem of relevant quantization has to be addressed directly, by preserving the relevant information about another variable.

- The relevance variable, denoted here by Y , must not be independent from the original signal X, namely they have positive mutual information $I(X;Y)$.

- It is assumed here that we have access to the joint distribution $p(x,y)$, which is part of the setup of the problem, similarly to $p(x)$ in the rate distortion case

The Information Bottleneck

- We would like our relevant quantization $\tilde{X}$ to compress X as much as possible.
- In contrast to the rate distortion problem, however, we now want this quantization to capture as much of the information about Y as possible.
- The amount of information about Y in X is given by

$$I(\tilde{X};Y) = \sum_{y}\sum_{\tilde{x}} p(y,\tilde{x}) \log \frac{p(y,\tilde{x})}{p(y)p(\tilde{x})} \leq I(X;Y).$$

- As with rate and distortion, there is a tradeoff between compressing the representation and preserving meaningful information.
- The assignment we are looking for is the one that keeps a fixed amount of meaningful information about the relevant signal Y while minimizing the number of bits from the original signal X (maximizing the compression).
- In effect we pass the information that X provides about Y through a "bottleneck" formed by the compact summaries $\tilde{X}$.

The Information Bottleneck

- Find the optimal assignment by minimizing the functional

$$\mathcal{L}[p(\tilde{x}|x)] = I(\tilde{X};X) - \beta I(\tilde{X};Y). \tag{15}$$

Theorem 4 *The optimal assignment, that minimizes Eq. (15), satisfies the equation*

$$p(\tilde{x}|x) = \frac{p(\tilde{x})}{Z(x,\beta)} \exp\left[-\beta \sum_y p(y|x) \log \frac{p(y|x)}{p(y|\tilde{x})}\right], \tag{16}$$

where the distribution $p(y|\tilde{x})$ in the exponent is given via Bayes' rule and the Markov chain condition $\tilde{X} \leftarrow X \leftarrow Y$, as,

$$p(y|\tilde{x}) = \frac{1}{p(\tilde{x})} \sum_x p(y|x)p(\tilde{x}|x)p(x). \tag{17}$$

$$p(\tilde{x}|x) = \frac{p(\tilde{x})}{Z(x,\beta)} \exp\left(-\beta D_{KL}\left[p(y|x)|p(y|\tilde{x})\right]\right) \qquad Z(x,\beta) = \exp[\beta\tilde{\lambda}(x)] = \sum_{\tilde{x}} p(\tilde{x}) \exp\left(-\beta D_{KL}\left[p(y|x)|p(y|\tilde{x})\right]\right)$$

The Information Bottleneck

minimize $\mathcal{F}[p(\tilde{x}|x)] = I(X;\tilde{X}) + \beta\langle d(x,\tilde{x})\rangle_{p(x,\tilde{x})}$

$$p(\tilde{x}|x) = \frac{p(\tilde{x})}{Z(x,\beta)} \exp\left[-\beta d(x,\tilde{x})\right] \ , \qquad (8)$$

minimize $\mathcal{L}[p(\tilde{x}|x)] = I(\tilde{X};X) - \beta I(\tilde{X};Y)$

$$p(\tilde{x}|x) = \frac{p(\tilde{x})}{Z(x,\beta)} \exp\left(-\beta D_{KL}\left[p(y|x)|p(y|\tilde{x})\right]\right)$$

The Kullback–Leibler divergence, $D_{KL}[p(y|x)|p(y|\tilde{x})]$, *emerged* as the relevant "effective distortion measure" from our variational principle but is not assumed otherwise anywhere! It is therefore natural to consider it as the "correct" distortion $d(x,\tilde{x}) = D_{KL}[p(y|x)|p(y|\tilde{x})]$ for quantization in the information bottleneck setting.

The information bottleneck iterative algorithm

Theorem 5 *The self consistent equations (18), (19), and (28), are satisfied simultaneously at the minima of the functional,*

$$\begin{aligned} \mathcal{F}\left[p(\tilde{x}|x); p(\tilde{x}); p(y|\tilde{x})\right] &= -\langle \log Z(x,\beta) \rangle_{p(x)} && (29) \\ &= I(X;\tilde{X}) + \beta \langle D_{KL}[p(y|x)|p(y|\tilde{x})] \rangle_{p(x,\tilde{x})} && (30) \end{aligned}$$

where the minimization is done independently *over the convex sets of the normalized distributions,* $\{p(\tilde{x})\}$ *and* $\{p(\tilde{x}|x)\}$ *and* $\{p(y|\tilde{x})\}$*. Namely,*

$$\min_{p(y|\tilde{x})} \min_{p(\tilde{x})} \min_{p(\tilde{x}|x)} \mathcal{F}\left[p(\tilde{x}|x); p(\tilde{x}); p(y|\tilde{x})\right] \ .$$

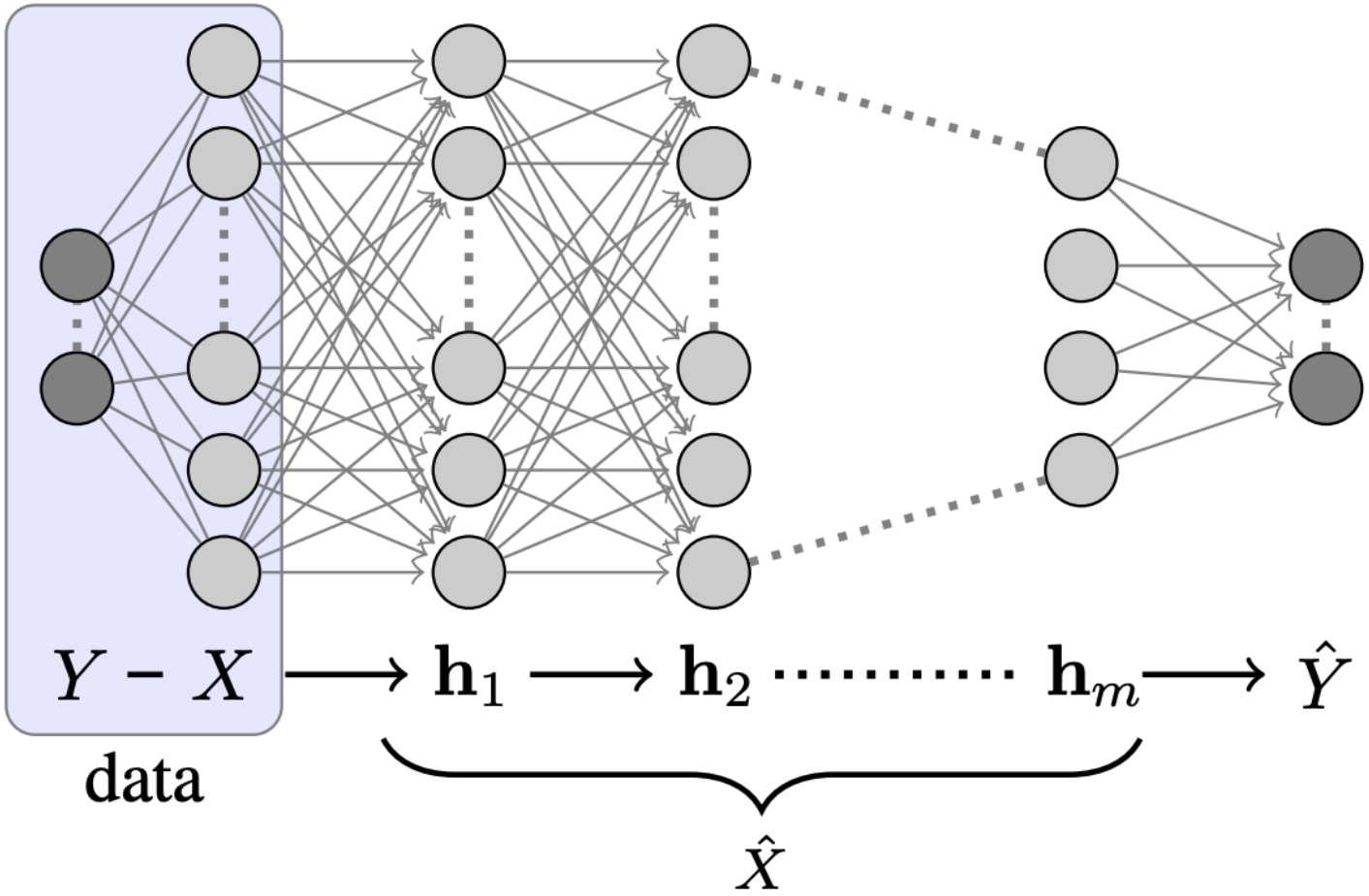
This minimization is performed by the converging alternating iterations. Denoting by t *the iteration step,*

$$\begin{cases} p_t(\tilde{x}|x) = \frac{p_t(\tilde{x})}{Z_t(x,\beta)} \exp(-\beta d(x,\tilde{x})) \\ p_{t+1}(\tilde{x}) = \sum_x p(x) p_t(\tilde{x}|x) \\ p_{t+1}(y|\tilde{x}) = \sum_y p(y|x) p_t(x|\tilde{x}) \end{cases} \qquad (31)$$

and the normalization (partition function) $Z_t(\beta, \tilde{x})$ *is evaluated for every* t *in Eq. (31).*

Deep Learning and the Information Bottleneck Principle [2015]

https://arxiv.org/pdf/1503.02406



https://www.youtube.com/watch?v=bLqJHjXihK8&feature=youtu.be

Opening the black box of Deep Neural Networks via Information [2017]

Opening the black box of Deep Neural Networks via Information

Ravid Schwartz-Ziv RAVID.ZIV@MAIL.HUJI.AC.IL
Edmond and Lilly Safra Center for Brain Sciences
The Hebrew University of Jerusalem
Jerusalem, 91904, Israel

Naftali Tishby* TISHBY@CS.HUJI.AC.IL
School of Engineering and Computer Science
and Edmond and Lilly Safra Center for Brain Sciences
The Hebrew University of Jerusalem
Jerusalem, 91904, Israel

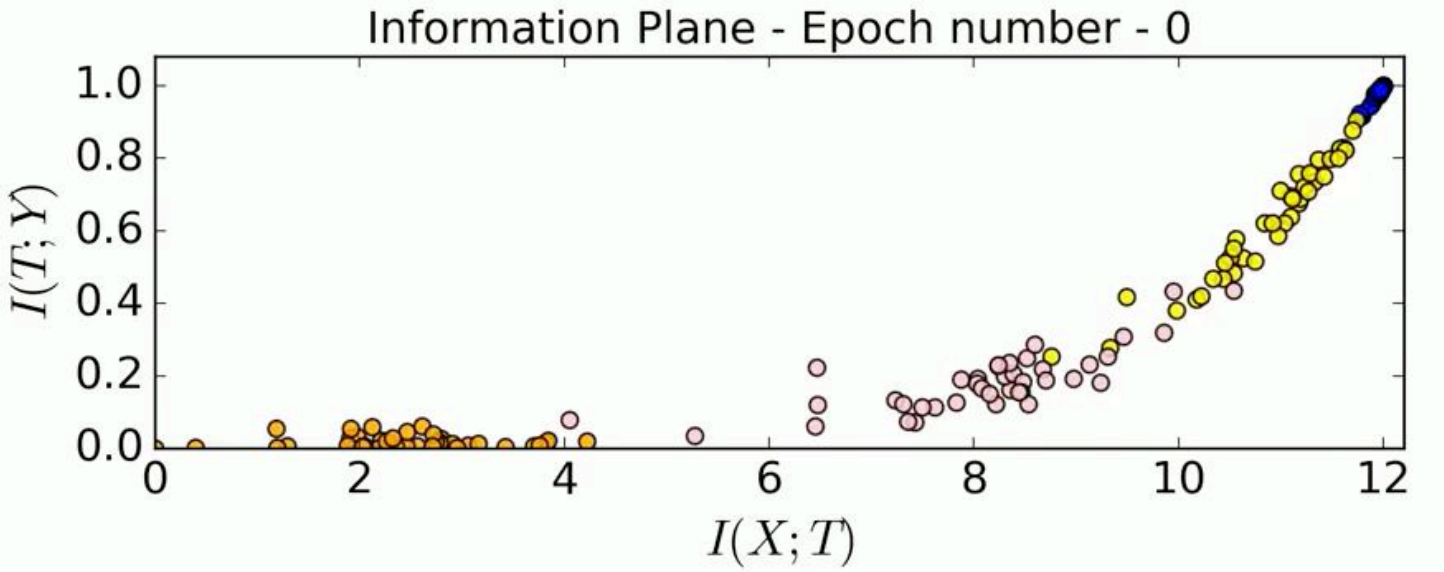
29 Apr 2017

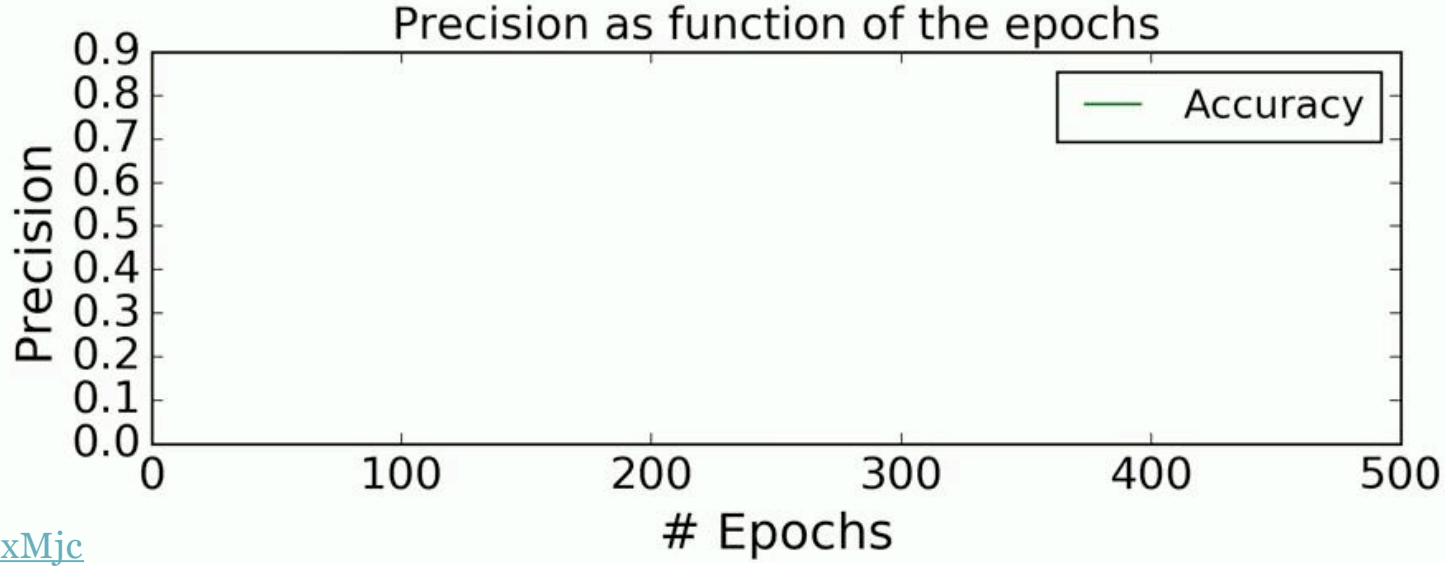
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https://arxiv.org/pdf/1703.00810

The information plane (Snapshots of layers)

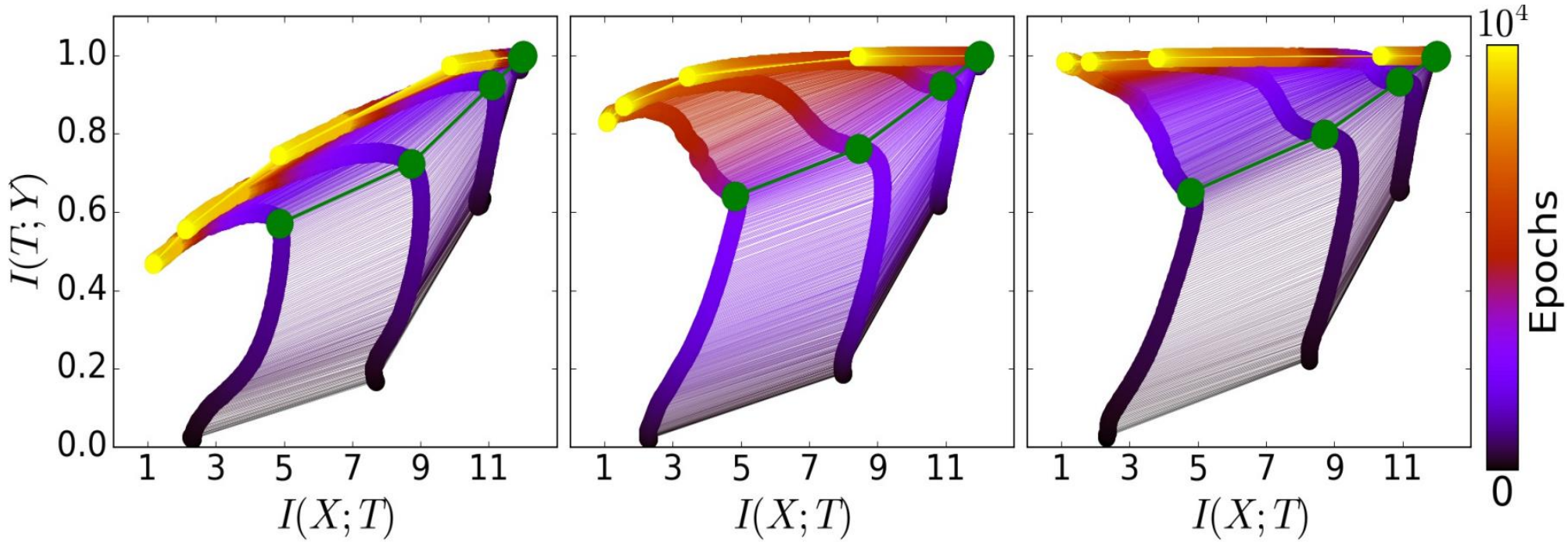
- Different layers are color-coders.
- **Green** being the layer *right next* to the input
- **Orange** being the *furthest*.





https://www.youtube.com/watch?v=P1A1yNsxMjc

The evolution of the layers with the training epochs in the information plane



- The evolution of the layers with the training epochs in the information plane, for different training samples. On the left - 5% of the data, middle - 45% of the data, and right - 85% of the data. The colors indicate the number of training epochs with Stochastic Gradient Descent from 0 to 10000. The network architecture was fully connected layers, with widths: input=12-10-8-6-4-2-1=output. The examples were generated by the spherical symmetric rule described in the text. The green paths correspond to the SGD drift-diffusion phase transition.

3. Why Language Models Hallucinate?

A paper [ICML 2026]

- 从**信息论率失真理论(Rate-Distortion)**角度证明：**幻觉是有限存储容量下，信息论最优策略必然带来的结果。**
- 针对“成员测试（membership testing）”任务：**即模型判断一条陈述是否属于真实事实集合，**解释为什么大模型即便数据完美、无训练 bug，**依然会产生高置信度幻觉。**

Hallucination is a Consequence of Space-Optimality: A Rate-Distortion Theorem for Membership Testing

Anxin Guo [1] Jingwei Li [2]

Abstract

Large language models often hallucinate with high confidence on “random facts” that lack inferable patterns. We formalize the memorization of such facts as a membership testing problem, unifying the discrete error metrics of Bloom filters with the continuous log-loss of LLMs. By analyzing this problem in the regime where facts are sparse in the universe of plausible claims, we establish a rate-distortion theorem: the optimal memory efficiency is characterized by the minimum KL divergence between score distributions on facts and non-facts. This theoretical framework provides a distinctive explanation for hallucination under an idealized setting: even with optimal training, perfect data, and a simplified “closed world” setting, the information-theoretically optimal strategy under limited capacity is not to abstain or forget, but to assign high confidence to some non-facts, resulting in hallucination. We validate this theory empirically on both synthetic and real-world data, showing that hallucinations persist as a natural consequence of lossy compression. The same theorem recovers and sharpens classical space lower bounds for Bloom-type filters, pinning down an additive constant left open for two-sided filters.

from a statistical learning perspective, by viewing an LLM as implementing a binary classifier over *random facts*—facts that are unstructured and cannot be logically inferred unless encountered during training, such as phone numbers or biographical details. By a no-free-lunch-style logic, since generalization is impossible on these facts, the model must make uninformed guesses on unseen random facts, leading to systematic inference errors. In the classification setting, hallucination corresponds to false positives resulting from such guesses.

As a natural workaround, researchers have promoted *abstention* (Kalai et al., 2025; Wen et al., 2025), where an LLM is expected to provide an “I don’t know” response on unseen random factual queries. Theoretically, we can further simplify the task by adopting a “closed-world assumption,” where all unseen potential facts are treated as non-facts at evaluation time [1]. Under this setting, an ideal LLM can be both informative and non-hallucinating on factual queries, as long as it accurately distinguishes the finitely many known facts from all other (potentially true but unknown) non-facts. This reduces hallucination mitigation to a sample-efficient memorization task.

However, empirical evidence suggests that LLMs struggle to perform this memorization task reliably. Even when permitted to abstain, models continue to generate high-confidence hallucinations and also exhibit “over-refusal” on legitimate

https://arxiv.org/pdf/2602.00906

大模型幻觉问题

- 现在的LLM做修改语法这样的有规律的任务已经效果不错了。
- 但是在一些随机事实上，即使最好的模型也很拉胯。
- 最著名的例子是有律师用GPT准备官司，结果GPT直接一个胡编乱造判例。
- 不允许上网搜索的话，询问LLM具体的论文标题和作者，得到的结果多半是似是而非的虚假论文。

The New York Times

Judge Fines Lawyers for MyPillow Founder for Error-Filled Court Filing

The judge said the lawyers had not explained how such errors could have been filed "absent the use of generative artificial intelligence or gross carelessness by counsel."

Listen · 5:24 min

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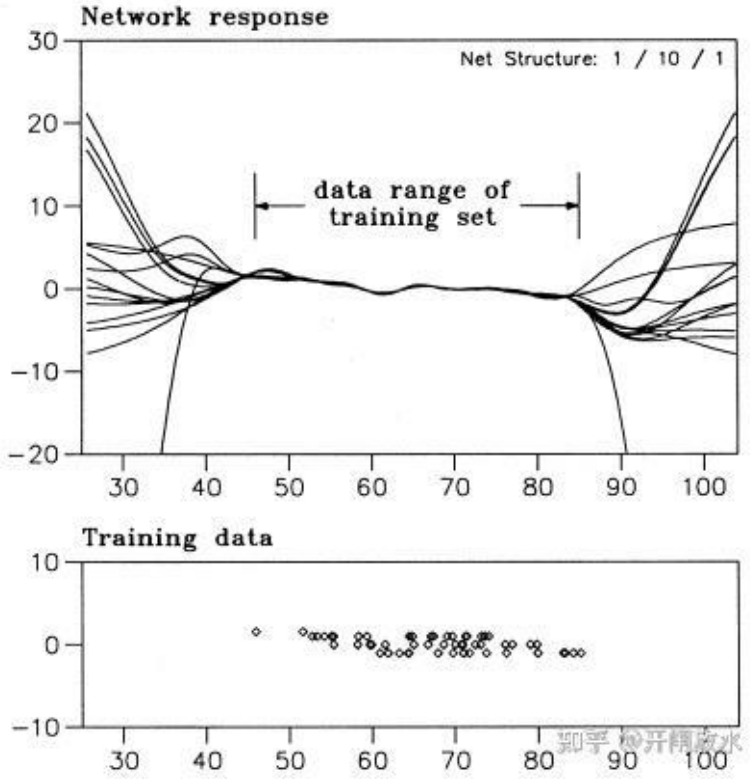


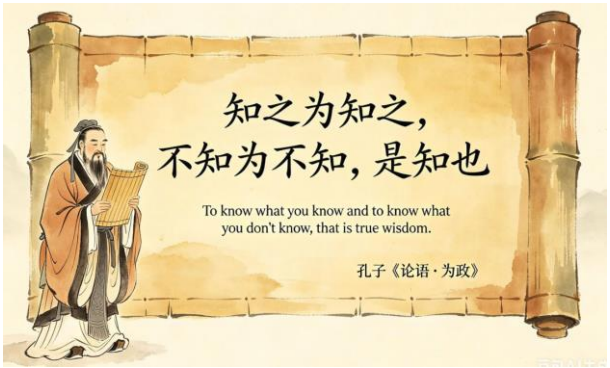
大模型幻觉的可能产生机制

- 工程上的问题
  - 数据脏、训练不足，或者是自回归本身就容易“不小心说错然后顺着往下说”
- 可计算性理论
  - 有些不可计算的输出LLM不可能完全学会，所以输出内容肯定会出错
- 统计学习理论
  - 在没有规律的死知识上，有的LLM压根没见过，被问到了只能瞎猜
- 模型容量/有损压缩
  - 模型大，但世界更大。训练集中的知识肯定有它记不住的

统计学习中经典的no-free-lunch定理

- 一些看起来随机的事实（例如电话号码），它们没有内在规律、也独立于其它知识，所以如果LLM想正确回答一个这样的问题，必须要在训练中见过才行。
- 假如在这些无规律的随机事实上进行真/假判别，那么我们除非训练中见过绝大多数随机事实，否则是不可能做全对的。
- 只要让LLM在没见过的、无法推断的问题上，直接拒绝回答(abstain)就行了。
- 很多训练和跑分并不奖励如实的承认自己不知道，这是当下幻觉产生的主要原因之一[1]。

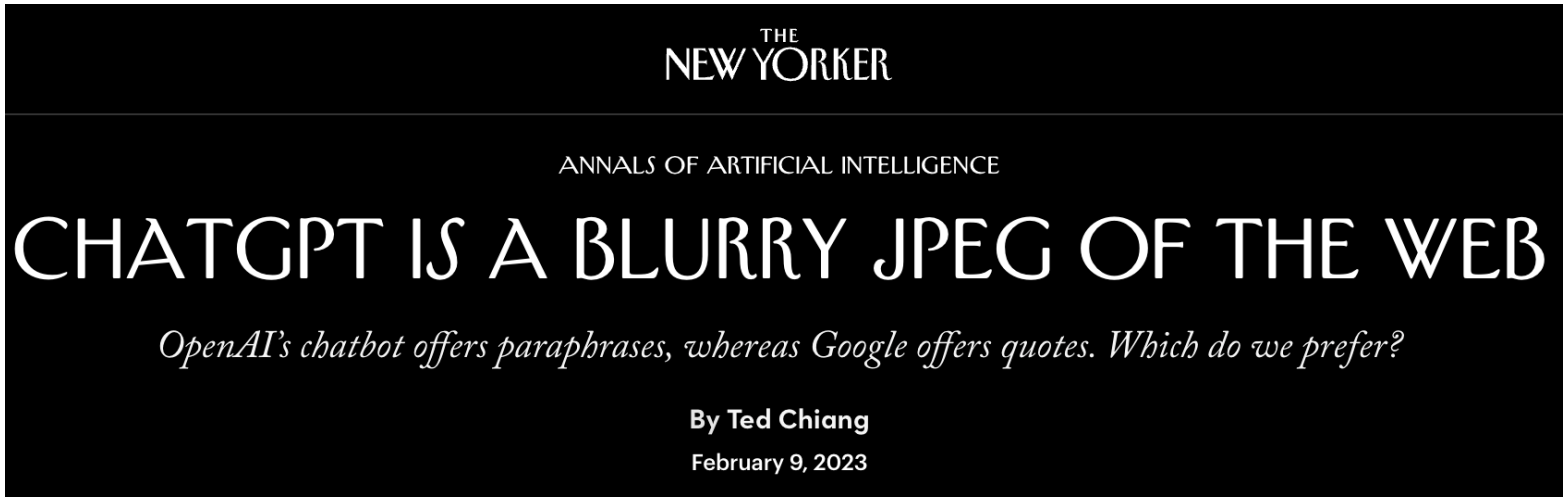




[1] Kalai, A. T., Nachum, O., Vempala, S. S., & Zhang, E. (2025). Why language models hallucinate. *arXiv preprint arXiv:2509.04664*.

有损压缩 (lossy compression)

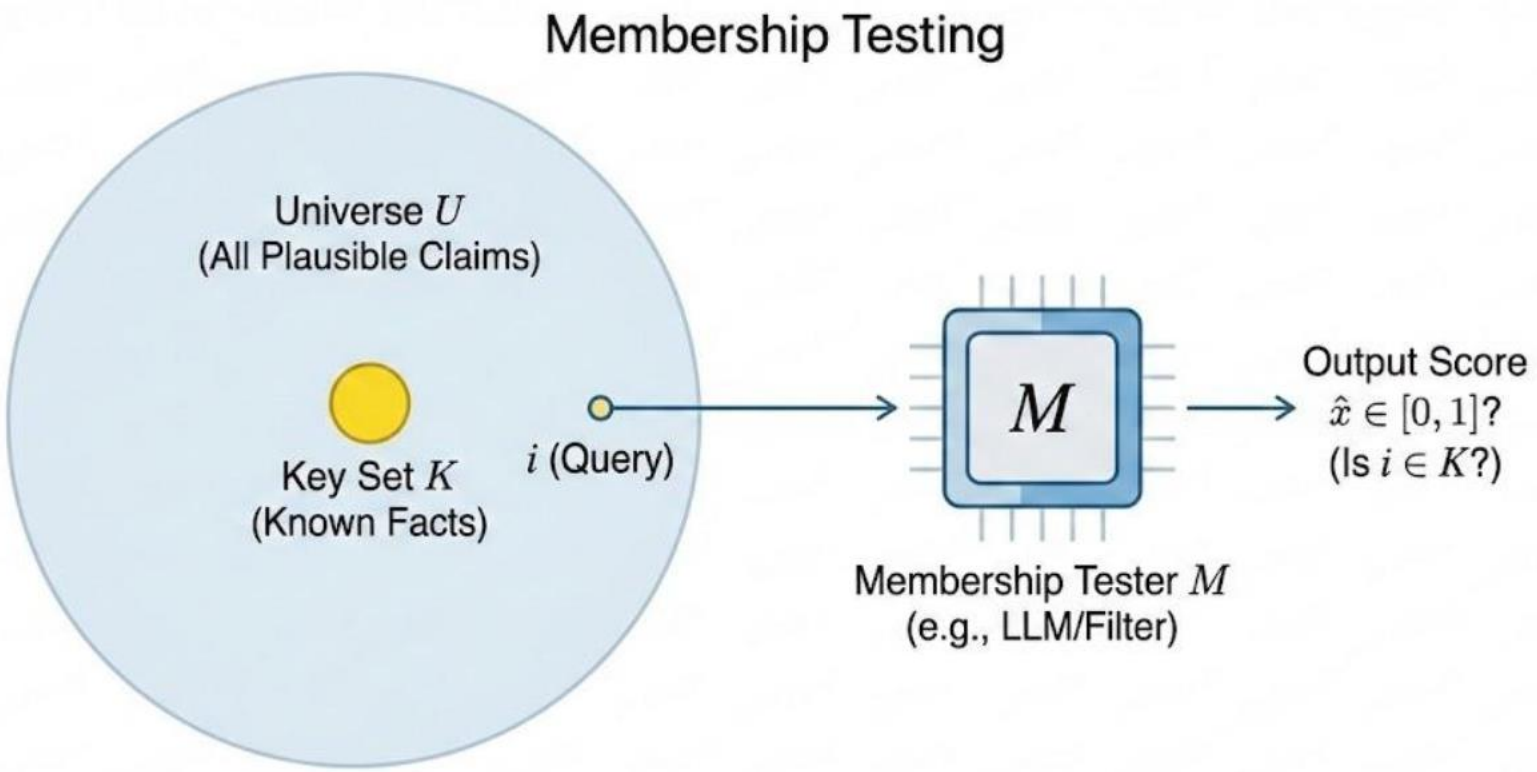
- LLM所包含的信息量不足以完整描述整个训练集，这就会导致细节信息的丢失。
- 在参数有限的时候，如果有无穷多的知识需要记忆，那些长尾的、不常见的知识，自然会被放弃。
- 两个谜团：（1）事实是有限的，为什么还是记不住？（2）记不住就算了，为什么非要瞎编？



https://arxiv.org/pdf/2602.00906

LLM是随机事实的“成员查询器” [ICML2026]

- LLM记忆知识时，它相当于构建了一个数据结构，这个结构要完成的任务很简单：
  - 使用者会查询一个元素，$i \in U$
  - 成员查询器会输出一个“置信度” $\hat{x}_i \in [0,1]$，代表它有多么确定 $i \in K$。换句话说，$\hat{x}_i$ 是 $\mathbb{I}\{i \in K\}$ 这个二元变量的近似。



A rate-distortion theorem

- 符号定义：
  - 宇宙 U ： 所有"看起来像事实"的潜在事实。
  - 事实集合 $K \subset U$ ： 模型真的见过、需要记住的那 n 条事实（随机事实场景下可以理解为：从 U 里均匀随机抽 n 个）。
  - 查询器的输出：对每个 $i \in U$ ，输出一个置信度 $\hat{x}_i \in [0,1]$ 。
  - μ_K ： 事实上的置信度分布，即 $\hat{x}_i$ 在 $i \in K$ 的时候的分布。
  - μ_N ： 非事实上的置信度分布，即 $\hat{x}_i$ 在 $i \notin K$ 的时候的分布。
- 主要定理：给定一个目标错误率，那么为了达到这个错误率，LLM（或任何“成员查询器”）充分必要的存储信息量，会由一个 KL 散度项决定。

想象宇宙里有一百万个“看起来像判例”的东西，但只有一千个是真的。对于查询器来说，默认状态下每个元素都大概率是非事实，因为事实太稀疏了。那么，想让查询器在遇到真事实时输出高置信度，它就必须为每一个事实额外存储一些信息，把它从默认的非事实中“拎出来”。这个"拎出来"所需的信息量，恰好就是事实和非事实的置信度分布之间的KL散度。KL散度越大，这个区分就越明显，但也越费空间。

A rate-distortion theorem

定理：在"事实稀疏"（$\frac{|K|}{|U|} \to 0$）的极限里，为了让错误率达到某个目标水平，**查询器需要存的总信息量大约是**

$$|K| \cdot \min_{\mu_K, \mu_N} \mathrm{KL}(\mu_K \| \mu_N) + o(|K|) \text{ bits},$$

其中 $\mathrm{KL}(\mu_K \| \mu_N)$ 就是*事实*和*非事实*上的置信度分布之间的KL 散度✦。并且，一个使用最小空间的成员查询器，其在事实和非事实上的**置信度分布**，在 $|K| \to \infty$ 的时候一定是**分别收敛到能够最小化这个KL 散度的** μ_K 和 μ_N 的。

对于熟悉信息论和统计的读者来说，这个结论非常自然：如果从非事实置信分布 μ_N 里采样 n 次，并且这些采样数据的empirical distribution"恰好看起来像 μ_K 来的"的概率，正好就是

$$\exp(-n \cdot \mathrm{KL}(\mu_K \| \mu_N) + o(n))$$

这正对应了我们上面说的直觉：所有元素默认都是非事实，所以**为了强行让事实上的置信度分布"像** μ_K **"而不是"像** μ_N **"，就得为了每一个事实，塞进一个KL 散度这么多额外信息。**

Thank you!